Case Study: Evaluating an AI System for HR Policy Questions

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Case Background

A company uses an AI-based system to answer common employee questions about HR policies. The system uses official company policy documents to find relevant information and prepare an answer.

Typical questions may include:

- How many days of annual leave are available?
- What is the process for applying for parental leave?
- Is medical documentation required for sick leave?
- Which employees are eligible for a particular benefit?

The system is expected to provide correct answers for routine questions. Questions that are unclear, sensitive, or difficult to answer from the available policy documents should be sent to an HR specialist.

The main purpose of the evaluation is to determine whether the system is accurate, reliable, fast, and cost-effective.

Theory

How the System Works

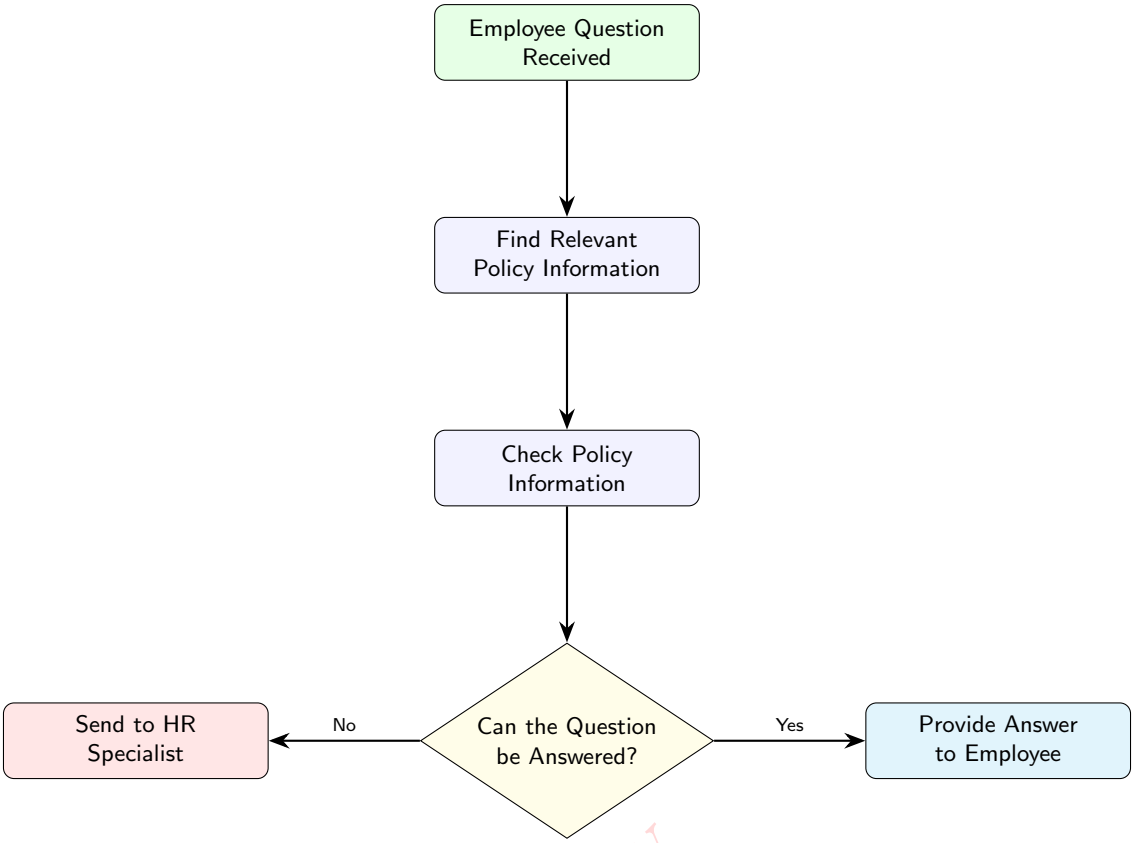
The HR policy system follows four main stages:

1. **Retrieve Information:** Find the relevant section of the official HR policy documents.
2. **Check the Information:** Use the retrieved policy information to understand the employee's question and determine the appropriate answer.
3. **Provide an Answer:** Give the employee a clear response based on the available policy information.
4. **Escalate When Necessary:** Send the question to an HR specialist when the question is too complex, unclear, or outside the available policy information.

For example, a question about the number of annual leave days may be answered directly when the policy clearly states the entitlement. A question involving an unusual employment situation may require review by an HR specialist.

System Flow

The following diagram shows the basic flow of the HR policy system:



Theory

Why the System Needs Evaluation

A correct answer is not the only measure of a useful HR system. A good system should:

- answer questions correctly;
- use information from the official policy documents;
- avoid unsupported or incorrect claims;
- complete routine questions without unnecessary HR intervention;
- provide answers within a reasonable time;
- operate at an acceptable cost; and
- remain reliable when the number of questions increases.

Evaluation should therefore use several measures rather than a single accuracy score.

Important Performance Measures

1. Task Success Rate

Task success rate shows the percentage of questions that are handled successfully.
Task Success Rate = Successful Tasks Total Tasks × 100
For example, if 184 out of 200 employee questions are successfully answered:
184 200 × 100 = 92
The task success rate is therefore 92

2. Failure Rate

Failure rate shows the percentage of questions that are not successfully completed.

Failure Rate = Failed Tasks Total Tasks × 100

If 184 out of 200 questions are successful, then:

200 − 184 =16

questions are unsuccessful.

Therefore:

16 200 × 100 = 8

The failure rate is 8

3. Factual Error or Hallucination Rate

An AI system may sometimes provide information that is incorrect or not supported by the official policy documents. Such an error can be measured as follows:

Hallucination Rate = Incorrect or Unsupported Claims Claims Checked × 100

For example, if 8 errors are found among 250 checked claims:

8 250 × 100 = 3.2

The hallucination rate is 3.2

4. Cost per Task

The cost of processing a question depends on the number of input and output tokens and the price per token.

Cost per Task = (Input Tokens × Input Price) + (Output Tokens × Output Price)

The result should be reported in the applicable currency.

5. Reliability of Multiple Steps

An AI system may need to complete several steps before providing an answer. For example, the system may need to retrieve a policy, identify the correct section, interpret the information, and prepare an answer.

If every step has approximately the same success rate, an approximate reliability for n sequential steps can be written as:

Chained Reliability ≈ (Single-Step Success Rate) n

This is an approximation. The calculation assumes that the steps have similar success rates and that the outcome of one step does not strongly affect another step.

Continuous Evaluation

The system should be evaluated regularly because performance can change after updates to the model, policies, prompts, retrieval methods, or system settings.

A simple evaluation cycle is:

Evaluate ⟶ Deploy ⟶ Monitor ⟶ Improve ⟶ Re-evaluate

For example, a drop in task success rate may indicate that the system needs better document retrieval or clearer instructions.

Theory

Practical Performance Scenarios

1. **Baseline Success Rate**

   An evaluation uses 200 employee questions. The system successfully resolves 184 questions.

   (a) Calculate the task success rate.

   (b) Calculate the number of unsuccessful questions.

   (c) Calculate the failure rate.

2. **Improving the Success Rate**

   The current task success rate is 92

   (a) How many successful answers are required to achieve 95

   (b) How many additional successful answers are needed compared with the current result of 184 successful answers?

3. **Checking Factual Accuracy**

   A quality check examines 250 policy claims produced by the system. Comparison with the official company handbook identifies 8 incorrect claims.

   Calculate the hallucination rate.

4. **Setting an Error Limit**

   The HR department sets a maximum hallucination rate of 2

   An evaluation checks 500 policy claims.

   What is the maximum number of incorrect claims allowed while remaining within the 2

5. **Calculating Processing Cost**

   A single transaction uses 2,000 input tokens and produces 500 output tokens.

   The input token price is | 0.001 per token, and the output token price is | 0.002 per token.

   Calculate the total cost of processing the transaction.

6. **Interpreting the Results**

   The following results are recorded during an evaluation:

   - Task success rate: 92
   - Failure rate: 8
   - Hallucination rate: 3.2
   - Maximum acceptable hallucination rate: 2

   Based on these results:

   (a) Which performance measure meets the target?

   (b) Which performance measure does not meet the target?

   (c) What improvement should be considered before the system is used for a larger number of employee questions?

Theory

Submission guidelines

- **Last Date: 21-10-2026**
- **Late Policy:** Late submissions face an automatic 10% point deduction per 24-hour period post-deadline.
- **Compilation Standards:** Documents that fail to compile due to unmapped Unicode or syntax errors will receive zero marks for presentation.